Situation-Aware Framework for Energy Saving in Unattended Activities of Daily Living in Smart Home Environments

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Goal

To propose a situation-aware framework that leverages a smart home resident's context information with respect to unattended ADLs in mitigating increased energy consumption in smart home environments.

An ADL: Basic self-care tasks performed on a daily basis, that are necessary for independent living

A smart home environment: An intelligent and inclusive residential space designed to instill comfort in a resident's daily living.



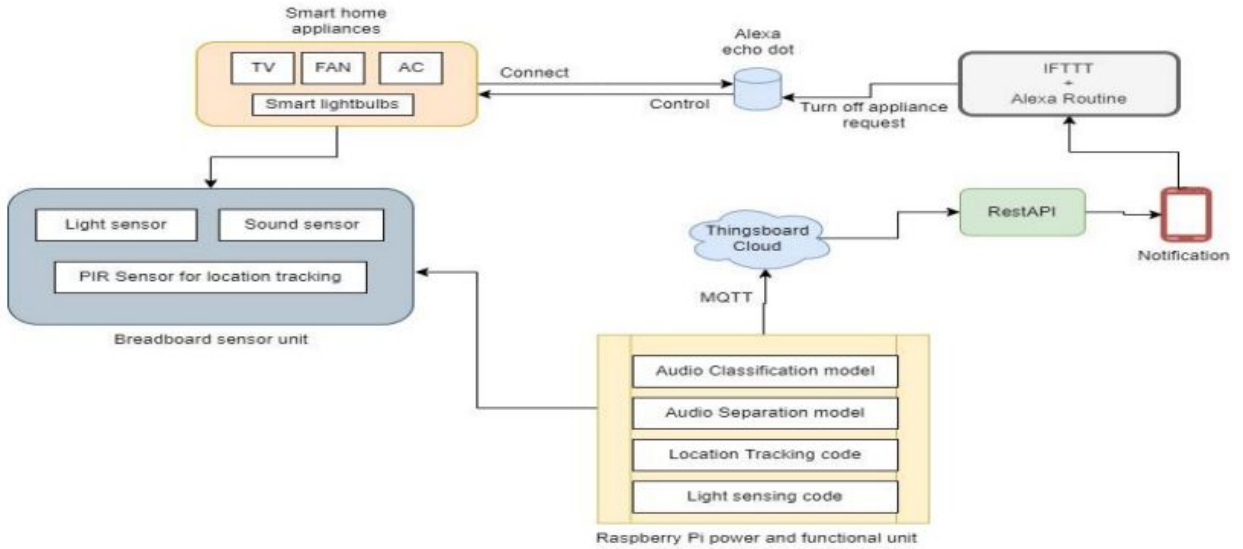
Introduction

- A Smart Home includes a network of sensors and IoT devices that can detect and monitor events, as well as adapt and respond to their needs.
- Several studies have shown a strong correlation between the amount of energy consumed in homes and how aging adults perform their ADLs
- To address this, our study presents a situation-aware framework for the mitigation of an increase in energy consumption from detected unattended/unsupervised ADLs in smart home environments.



Awareness and Situation Aware Framework

- Impaired Awareness: Causes aging adults to underestimate their functional decline and can have economic consequences and compromise their safety.
- Situation Awareness: Situation awareness refers to the ability to perceive or sense devices in the environment at a given time instance and location, understand their operational context, and predict their future states or status.



Description of Framework and Components

The smart devices are connected and controlled through the Alexa Echo Dot.

The Raspberry Pi serves as the control hub of the smart home environment, detecting the presence of light and sound through its connected sensors.

This helps to detect the presence of a resident within a space. If no presence is detected in space, it sends a notification to the resident's phone instructing them to turn off any detected devices that are left unattended or not turned off.

The Alexa Echo Dot can receive requests initiated by the user from our custom-built mobile app, allowing them to turn off the smart devices when they receive a notification on their phone from the Raspberry Pi.

DashHome Audio Dataset

- 60-second audio recordings of the following devices: AC, Fan, TV, and heater
- Consists of 712 samples in .wav format
- The recordings were collected in a quiet room with a recording device at a sample rate of 44kHz and then later re-sampled to 16kHz.

Devices	Parameters
AC	Temperature, Speed, Sensor level
Fan	Speed, Sensor level
TV	Music/ Spoken sound, Volume, Sensor level
Heater	Speed, Sensor level



Light Control



Light sensors sense the lighting emitted from the lightbulbs.



Location sensors detect human presence within the room.



If absence is detected, the user receives a notification for turning off the lightbulb.

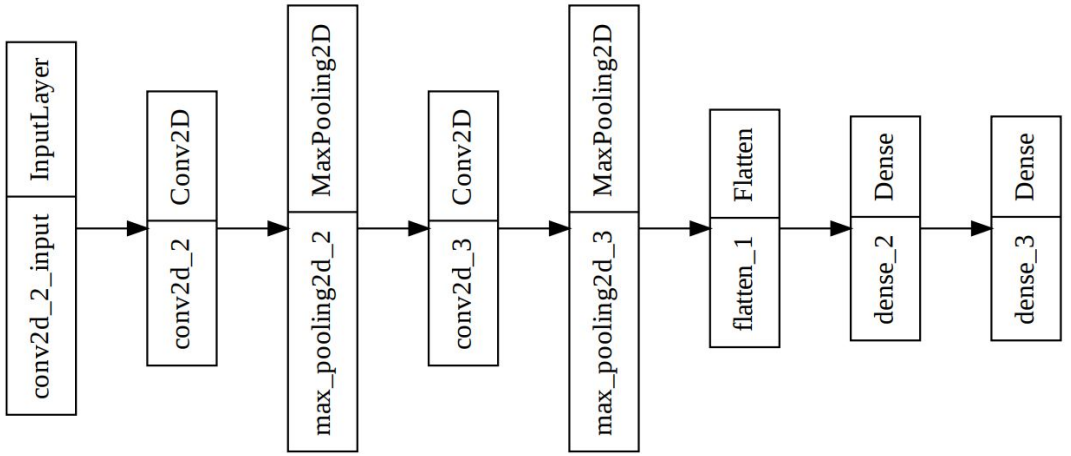
Audio Source Separation

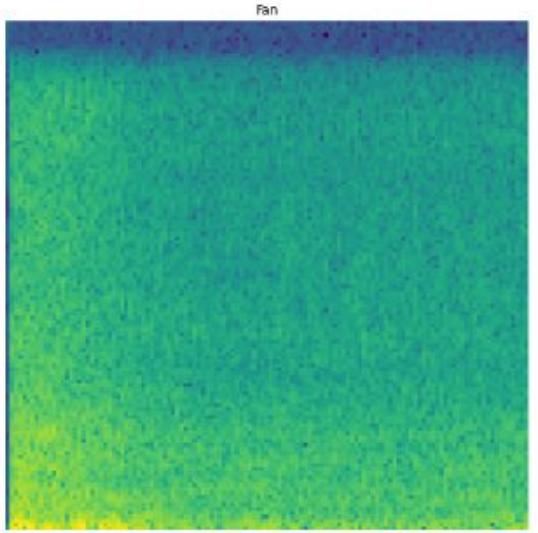
- Blind source separation (BSS) refers to receiving a set of source signals from a mixed signal or a set of mixed signals with very little information of the corresponding source signals
- We employ four different variations of Independent Vector Analysis (IVA) in our study: NaturalGradLaplaceIVA, GradLaplaceIVA, AuxLaplaceIVA, AuxGuassIVA
- In our experiment, we assume the functioning of two corresponding appliances at the same time.
- Other hyperparameters in the algorithm include: the presence of noise, spatial algorithm parameters, projection back parameter

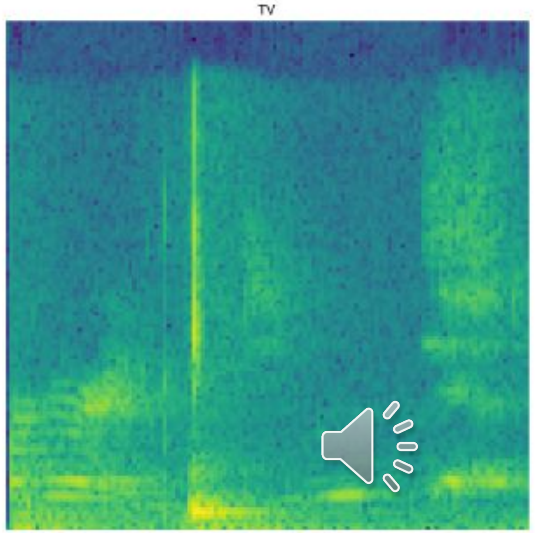


Audio Classification

- Audio classification is done by conversion of the .wav files into two forms: Short Time Fourier Transform (STFT) and mel-spectrograms.
- We take the spectrogram approach. Once we receive the spectrogram, the audio classification problem has been changed to a computer vision problem.
- We employ a robust CNN network for the feature extraction of the spectrogram.







BSSP Formula and Corresponding Output

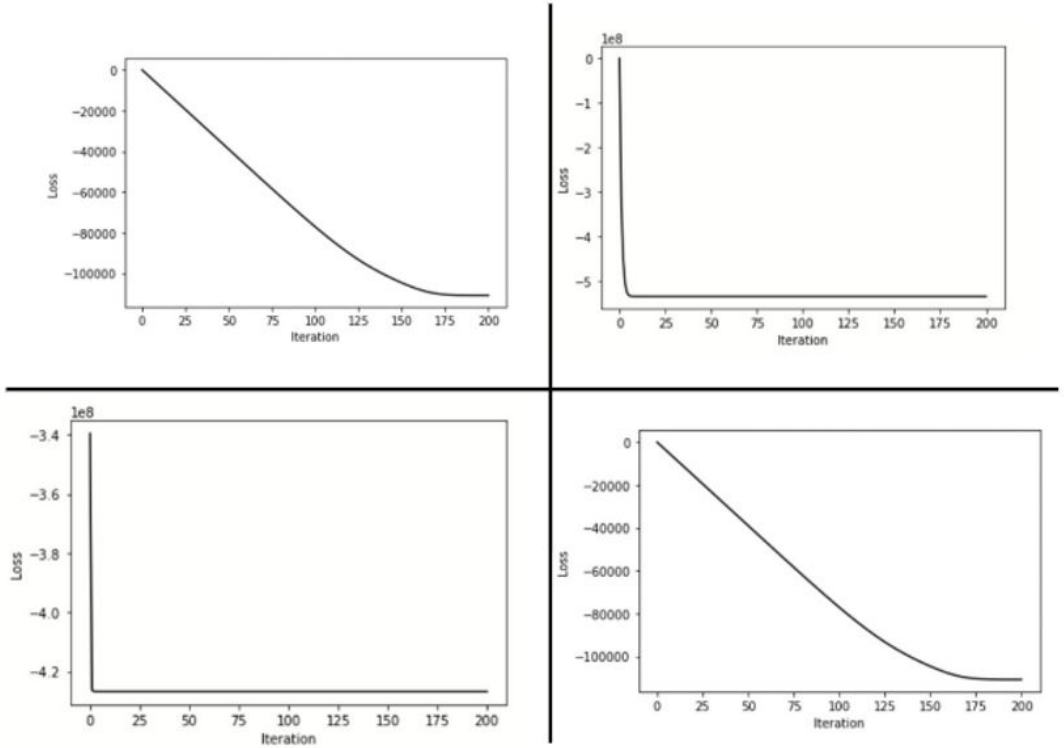
- The BSSP formula is the Blind Source Separation Probabilistic formula.
- It is used for calculating the net probability of the separated outputs factoring in the total outputs and number of true predicted outputs from the audio separator and classifier

TV + Fan	TV & Fan	TV x 6, Fan x 4, TV, Fan, TV, Fan, TV x 8	0.77
TV + AC	Fan & TV	Fan, TV, Fan, TV, Fan, TV, Fan x 4, Heater, TV x 3, Fan, TV, Fan, TV, Fan, TV, Fan, TV	0.45
Fan + AC + TV	Fan & AC	TV, Fan, TV, Fan, TV, Fan x 8, AC x 2, Fan, AC, Fan, AC, Fan, AC, Fan	0.69

The BSSP Formula.

$$p = \sum_{i=1}^{n} \frac{min\left(\frac{N}{i}, x_i\right)}{N}$$

Model Comparisons



- Compared to Laplacian, the Gaussian model generally has a much-refined output and higher probability.
- The Auxiliary models are independent of iterations after a certain number
- Models trained on TV and fan will never give an AC output, irrespective of the iterations and model.
- The Ag_iss_TVAC is the only model that could detect the AC as AC, hence it performed better in all scenarios. Training it over the AC dataset makes the difference that gives us the AC as output instead of the Fan.

Energy Savings Outcome

- To provide a more specific estimate of energy savings, we can calculate the energy consumed every 30 minutes or hour based on the power consumption
- Formula used: $E = P_a * T_s - P_m * T$

Ground truth (Smart devices)	Power rating of each device/combination of devices (Watt)	Energy saved in unattended ADL for a duration of 1 hour (W-hour)	Energy saved in unattended ADL for a duration of 30 minutes(W-hour)
AC	1470	1466	733
Fan	54.7	50.7	25.35
TV	219	215	107.5
Fan + AC	1524.7	1520.7	760.35
TV + Fan	273.7	269.7	134.85
TV + AC	1689	1685	84.25
Fan + AC + TV	1743.7	1739.7	869.85



Conclusion and Future Work

- In this study, we proposed a situation-aware framework for mitigating unintended energy consumption in unattended ADLs.
- The framework includes low-cost sensors and IoT devices such as Raspberry Pi, sound, light and motion sensors.
- In addition, the framework incorporates machine learning models that use robust context data generated from the smart home appliances to enable the user to understand their ADLs environment, perceive smart home appliances and their operating states to mitigate unintended energy consumption when ADLs are unattended.
- For our future work, we plan to test the efficacy of our proposed framework in senior retirement homes in using longitudinal study.



Thank you

